

Project Progress Report — MSML640 (Computer Vision)

Project Title: AU-Conditioned MAE + Pose-Normalized FER

Course: MSML640 — Computer Vision

Group Members: Narendra Rane | Prakhar Tiwari

Submission Date: October 22, 2025

1. Overview

This project aims to enhance **Facial Expression Recognition (FER)** by combining **pose normalization** with **Action Unit (AU)-conditioned Masked Autoencoder (MAE)** pretraining. Narendra Rane's individual focus includes implementing the **Pose Normalization module**, preparing datasets, and performing **ablation experiments** to compare baseline and pose-aware models. Prakhar Tiwari's individual focus includes AU-MAE pretraining which is a work in progress.

The system architecture follows a two-stage pipeline:

1. **Pose Normalizer (STN)** for frontalization and pose correction.
 2. **Encoder + FER Head** fine-tuned on emotion classes.
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2. Work Completed

a. Dataset Preparation

- Downloaded and organized the RAF-DB dataset (<https://www.kaggle.com/datasets/shuvoalok/raf-db-dataset>).
- Created training and validation splits using `make_raf_splits_from_folders.py`.
- Verified image counts (12,271 train / 3,068 val) and labels for consistency.

b. Baseline FER Implementation (Ablation A)

- Implemented a ResNet-50 backbone FER model.
- Trained for 10 epochs on RAF-DB (Adam optimizer, batch size 64).
- Achieved **81.45 % accuracy**, **69.20 % balanced accuracy**, **0.717 macro-F1**.

c. Pose Normalization Module (Ablation B)

- Integrated a lightweight **Spatial Transformer Network (STN)** with BatchNorm and Dropout for pose correction.

- Applied warm-up training to stabilize learning and prevent gradient explosion.
- Achieved **79.95 % accuracy**, **68.26 % balanced accuracy**, **0.707 macro-F1** after 10 epochs.

Experiment	Accuracy (%)	Balanced Accuracy (%)	Macro F1
Ablation A (No Pose Normalizer)	81.45	69.20	0.717
Ablation B (With Pose Normalizer)	79.95	68.26	0.707

3. Observations

- Baseline FER model shows strong and stable performance.
 - Pose Normalizer successfully aligns faces without destabilizing training.
 - Slight performance drop indicates need for longer fine-tuning and adaptive learning-rate schedules.
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4. Challenges

- **Training Time:** Local Apple M3 Pro (MPS) GPU limits batch size and runtime speed.
 - **Pose Normalizer Instability:** Initially unstable in early epochs; solved with warm-up training and lower LR.
 - **Joint Optimization:** Balancing FER and pose losses requires careful scheduler tuning.
 - **AU-Images:** for AU-MAE we require to modify a dataset to associate aus with each image.
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5. Next Steps

- Extend Pose Normalizer training to **20 epochs** with adaptive learning rate decay.
 - Integrate the AU-Conditioned MAE pretraining on AffectNet to enhance feature robustness.
 - Validate on **FER-2013** and **AffectNet** for cross-dataset generalization.
 - Conduct “data-in-the-wild” evaluation to assess robustness under occlusions and extreme poses.
 - Generate accuracy/loss plots and confusion matrices for the final report.
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6. Proof of Progress

a. Training Output (Kaggle Results)

Below are the outputs from the Kaggle training runs for both **Ablation A (Baseline FER)** and **Ablation B (FER + Pose Normalizer)**.

Ablation A — FER Baseline (No Pose Normalization)

```
[Ablation A] FER baseline (no pose normalization)
Epoch 1/10 | loss=0.9886
Epoch 2/10 | loss=0.5613
Epoch 3/10 | loss=0.3842
Epoch 4/10 | loss=0.2849
Epoch 5/10 | loss=0.2120
Epoch 6/10 | loss=0.1717
Epoch 7/10 | loss=0.1277
Epoch 8/10 | loss=0.1074
Epoch 9/10 | loss=0.0977
Epoch 10/10 | loss=0.0911
Ablation A metrics: {'accuracy': 0.8145371577574967, 'balanced_accuracy': 0.6920394280130054, 'macro_f1': 0.7170668620967982, 'confusion_matrix': array([[ 230,   17,    3,   13,   16,    3,   47],
        [  11,   38,    1,    6,   11,    3,    4],
        [   1,    4,   55,   11,   37,    5,   47],
        [   4,    1,    4, 1066,   26,    4,   80],
        [   1,    0,    2,   14,  422,    3,   36],
        [   3,    3,    8,   18,   12,  105,   13],
        [   6,    2,    3,   16,   69,    1,  583]])}
```

→ Shows 10-epoch training log with steadily decreasing loss and stable validation accuracy.

→ Final metrics:

- Accuracy: **81.45 %**
- Balanced Accuracy: **69.20 %**
- Macro F1: **0.717**

Ablation B — FER + Pose Normalizer

[Ablation B] FER + PoseNormalizer

[Warmup] Training PoseNormalizer only for 1 epoch

Warmup PoseNormalizer: 100%|██████████| 192/192 [00:23<00:00, 8.07it/s]

[✓] Warmup complete — PoseNormalizer stabilized.

```
Epoch 1/10 | loss=1.9279
Epoch 2/10 | loss=1.9279
Epoch 3/10 | loss=1.5283
Epoch 4/10 | loss=1.1008
Epoch 5/10 | loss=0.8315
Epoch 6/10 | loss=0.6710
Epoch 7/10 | loss=0.5307
Epoch 8/10 | loss=0.4192
Epoch 9/10 | loss=0.3448
Epoch 10/10 | loss=0.2592
```

```
Ablation B metrics: {'accuracy': 0.7995436766623207, 'balanced_accuracy': 0.6825535102947933, 'macro_f1': 0.7068393656513601, 'confusion_matrix': array([[ 256,    9,    6,   10,   10,    3,   35],
       [ 15,   36,    1,    7,    9,    3,    3],
       [  1,    1,   65,   21,   19,   10,   43],
       [ 10,    0,   17, 1076,   27,    3,   52],
       [  4,    1,   15,   29,  353,    4,   72],
       [  9,    5,   11,   15,    5,  102,   15],
       [ 12,    0,   21,   36,   46,    0,  565]])}
```

→ Includes warm-up stage for stabilizing PoseNormalizer.

→ Final metrics:

- Accuracy: **79.95 %**
- Balanced Accuracy: **68.26 %**
- Macro F1: **0.707**

Ablation Summary

=== Ablation Summary ===

```
no_pose {'accuracy': 0.8145, 'balanced_accuracy': 0.692, 'macro_f1': 0.7171, 'confusion_matrix': '\n[[ 230  17   3  13  16   3  47]\n[ 11  38   1   6  11   3   4]\n[   1   4  55   1  37   5  47]\n[   4   1   4 1066  26   4  80]\n[   1   0   2  14 422   3  36]\n[   3   3   8  18  12 105 13]\n[   6   2   3  16  69   1 583]]'}\nwith_pose {'accuracy': 0.7995, 'balanced_accuracy': 0.6826, 'macro_f1': 0.7068, 'confusion_matrix': '\n[[ 256   9   6  10  10   3  35]\n[ 15  36   1   7   9   3   3]\n[   1   1  65  21  19  10  43]\n[ 10   0  17 1076  27   3  52]\n[   4   1  15  29 353   4  72]\n[   9   5  11  15   5 102  15]\n[ 12   0  21  36  46   0 565]]'}
```

→ Displays direct comparison between the two configurations confirming consistent integration of pose normalization.

7. GitHub Activity Snapshot

Our GitHub repository demonstrates consistent progress and individual contributions throughout the project development cycle. The following snapshots capture the commit history and recent updates:

Repository: https://github.com/narendrarane50/MSML640-Final_Project

Commit Activity Proof (Screenshots):

- Screenshot 1: Main Branch commit history

Commits		
main	All users	All time
Commits on Oct 22, 2025		
updated readme file	narendrarane50 authored 4 hours ago	Verified 3a03ae0
Commits on Oct 21, 2025		
FER baseline and PoseNormalizer	Narendra Rajendra Rane authored and Narendra Rajendra Rane committed yesterday	c0cb04a
Commits on Oct 19, 2025		
added some temp files	Narendra Rajendra Rane authored and Narendra Rajendra Rane committed 3 days ago	8ad28a9
architecture of project	Narendra Rajendra Rane authored and Narendra Rajendra Rane committed 3 days ago	d23823b
Added Project Proposal file	peter18775 committed 4 days ago	e353fdb
updated Readme file	peter18775 authored 4 days ago	Verified 8a1c518
Added Readme file	peter18775 committed 4 days ago	a089dc6

- Screenshot 2: prakhar_tiwari branch commit history

Commits		
prakhar_tiwari	All users	All time
Commits on Oct 22, 2025		
Train_MAE without AU for Baseline	peter18775 committed 37 minutes ago	fb325f9
AU_MAE-code written; util files added	peter18775 committed 5 hours ago	2be6062
Commits on Oct 21, 2025		
FER baseline and PoseNormalizer	Narendra Rajendra Rane authored and Narendra Rajendra Rane committed yesterday	c0cb04a
Commits on Oct 19, 2025		
added some temp files	Narendra Rajendra Rane authored and Narendra Rajendra Rane committed 3 days ago	8ad28a9
architecture of project	Narendra Rajendra Rane authored and Narendra Rajendra Rane committed 3 days ago	d23823b
Added Project Proposal file	peter18775 committed 4 days ago	e353fdb
updated Readme file	peter18775 authored 4 days ago	Verified 8a1c518
Added Readme file	peter18775 committed 4 days ago	a089dc6

Each commit clearly documents the development timeline, confirming active collaboration and adherence to project milestones.